

Module5_Lab2

October 3, 2026

```
[1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("custdata.tsv", sep="\t")

df.head()
```

```
[1]:   custid sex is.employed  income  marital.stat  health.ins \
0    2068  F         NaN   11300         Married         True
1    2073  F         NaN     0         Married         True
2    2848  M         True   4500  Never Married        False
3    5641  M         True  20000  Never Married        False
4    6369  F         True  12000  Never Married         True

      housing.type recent.move  num.vehicles  age state.of.res
0  Homeowner free and clear      False         2.0  49.0  Michigan
1              Rented          True         3.0  40.0   Florida
2              Rented          True         3.0  22.0   Georgia
3  Occupied with no rent      False         0.0  22.0 New Mexico
4              Rented          True         1.0  31.0   Florida
```

```
[2]: df.shape
```

```
[2]: (1000, 11)
```

```
[3]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   custid          1000 non-null  int64
1   sex             1000 non-null  str
2   is.employed     672 non-null   object
3   income          1000 non-null  int64
4   marital.stat    1000 non-null  str
5   health.ins      1000 non-null  bool
6   housing.type    944 non-null   str
```

```

7  recent.move    944 non-null    object
8  num.vehicles   944 non-null    float64
9  age            1000 non-null   float64
10 state.of.res   1000 non-null    str
dtypes: bool(1), float64(2), int64(2), object(2), str(4)
memory usage: 115.8+ KB

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```
[4]: df.describe()
```

```

[4]:
count    1.000000e+03    1000.000000    944.000000    1000.000000
mean     6.984997e+05    53504.771000     1.916314     51.699815
std      4.135083e+05    65478.065729     1.101618     18.863433
min      2.068000e+03    -8700.000000     0.000000     0.000000
25%      3.456668e+05    14600.000000     1.000000     38.000000
50%      6.934030e+05    35000.000000     2.000000     50.000000
75%      1.044606e+06    67000.000000     2.000000     64.000000
max      1.414286e+06    615000.000000     6.000000    146.680197

```

```
[5]: df.isnull().sum()
```

```

[5]: custid          0
sex                0
is.employed       328
income            0
marital.stat      0
health.ins        0
housing.type      56
recent.move       56
num.vehicles       56
age               0
state.of.res      0
dtype: int64

```

```

[6]: housing_counts = df['housing.type'].value_counts(dropna=True)

plt.figure
bars = plt.bar(
    housing_counts.index,
    housing_counts.values,
)

plt.title('Distribution of Housing Type (Excluding NA)')
plt.xlabel('Housing Type')
plt.ylabel('Number of Customers')
plt.show()

```

Cell In[6], line 9

```
plt.title('Distribution of Housing Type (Excluding NA))
^
SyntaxError: unterminated string literal (detected at line 9)
```

```
[7]: housing_counts = df['housing.type'].value_counts(dropna=True)

plt.figure
bars = plt.bar(
    housing_counts.index,
    housing_counts.values,
)

plt.title('Distribution of Housing Type (Excluding NA'))
plt.xlabel('Housing Type')
plt.ylabel('Number of Customers')
plt.show()
```

Cell In[7], line 9

```
plt.title('Distribution of Housing Type (Excluding NA'))
^
SyntaxError: unmatched ')'
```

```
[8]: housing_counts = df['housing.type'].value_counts(dropna=True)

plt.figure
bars = plt.bar(
    housing_counts.index,
    housing_counts.values,
)

plt.title('Distribution of Housing Type (Excluding NA)')
plt.xlabel('Housing Type')
plt.ylabel('Number of Customers')
plt.show()
```

